



THE GRANARY YOU FILL BEFORE THE DROUGHT

Hozi

National Food-Security Foresight Engine

Fill the granary before the drought.

Hozi helps the people who plan Zimbabwe's food supply see hunger coming months before it arrives, and test where help works best, so limited resources reach the districts that need them first. It turns scattered field signals into one clear, honest picture, early enough to act.

Submitted by

Curious Inq — AI-powered creative studio, Harare
Nash Barara · Product & Design Lead | [Technical Lead] | [Food-Security Advisor]

June 2026

Zimbabwe
Open-source · built locally

THE PROBLEM

We see the hunger coming. We just see it too late.

Almost every year, parts of Zimbabwe slide into food shortage when the rains fail and harvests fall short. The tragedy is rarely that no one saw it coming. It is that the warning arrives too late, buried in slow seasonal reports, after the season is already lost and scarce resources have been spent in the wrong places.

Today's food-security assessments are periodic, document-heavy, and often produced outside the country. By the time a district is flagged, the window to plant differently, move seed, or prepare relief has usually closed. Planners are asked to make multi-million-dollar decisions with a rear-view mirror.

THE SOLUTION

Hozì — a living early-warning engine for food security.

Hozì reads the signals that decide a harvest — rainfall, how green the crops are, pests, irrigation, and access to seed and fertiliser — for every district. It projects where food risk is heading over the next one to three months, and lets a planner test interventions *before* spending a cent.

WHO IT IS FOR

Hozì is not a farming app for individuals. It is a decision-support tool for the people who allocate resources: the Ministry of Lands & Agriculture, food-security and civil-protection officers, ZimVAC, and relief NGOs. It is the national "control room" view, not the single-field view.

HOW IT WORKS — FIVE CLEAR STEPS

1

Read signals

Rainfall, vegetation, pests, irrigation & input access, district by district.

2

Score risk

A transparent, explainable food-security risk score (0–100) per district.

3

Look ahead

Project the trend 1–3 months forward with an honest confidence range.

4

Test fixes

Simulate seed, irrigation or input support and see the risk drop.

5

Explain

Plain-language briefs and a clear map, in English & local languages.

In one line: Zimbabwe's food-security foresight runs on slow, foreign, once-a-season reports. Hozì makes it live, interactive, and ours — so we act before the crisis, not after it.

TECHNICAL SPECIFICATION

A transparent engine — not a black box.

Hoji is built so that every number it shows can be traced and explained. That matters when public money and public trust are on the line.

HOW IT IS BUILT

- **Data layer:** district-level climate, crop and market signals (rainfall, NDVI vegetation index, pests, irrigation, input access, prices, yields).
- **Risk engine:** a clear weighted model that converts those signals into a 0–100 food-security risk for each district.
- **Forecast:** seasonal trend projection 1–3 months ahead, with a widening confidence band the further out it looks.
- **Simulator:** adjust irrigation, inputs or seed support and the projected risk recalculates instantly.
- **Interface:** a web dashboard — national map, time-slider, district drill-down — that works on a normal browser.

PROVEN AGAINST THE DATA

Before trusting a forecast, you test the model. Hoji's risk score was validated against an independent agricultural risk index for the same districts.

0.81

CORRELATION BETWEEN HOZI'S RISK SCORE AND THE INDEPENDENT INDEX

A strong match means the model reflects real conditions on the ground — it is not guessing. From there, Hoji projects the same well-understood seasonal trend forward.

INNOVATIVE FEATURES

What makes Hoji different

Foresight, not hindsight.

It forecasts where risk is heading, instead of reporting where it already is.

A "what-if" simulator.

Planners test interventions and see the payoff before committing budget — so help goes where it saves the most.

Speaks our languages.

Early-warning briefs in English, chiShona and isiNdebele for the last mile.

Honest by design.

It states clearly what it can and cannot know. A visible "limitations" panel builds trust instead of overclaiming.

Sovereign & open.

Built in Zimbabwe, on Zimbabwean data, released open-source — not a foreign black box.

Sits on top of what exists.

It can use signals from existing services and the national data platform, rather than competing with them.

Responsible AI: Hoji is decision-support, not an oracle. It keeps a human expert in the loop, shows its workings, and is honest about uncertainty — in line with the ethical principles of the National AI Strategy.

PROJECTED BENEFITS

Acting one season earlier changes everything.

In a working test on national sample data, Hoji flagged a sharp seasonal rise in food-security risk — and showed exactly where it would land first.

0 → 19

DISTRICTS (OF 20) PROJECTED TO ENTER HIGH RISK BY SEPTEMBER WITHOUT ACTION

3 mo.

EARLIER WARNING THAN A TYPICAL SEASONAL REPORT CYCLE

1 map

REPLACES SCATTERED SPREADSHEETS WITH ONE SHARED, LIVE PICTURE

- **Fewer hungry families:** earlier warning means seed, irrigation and relief move *before* the harvest fails, not after.
- **More from every dollar:** the simulator directs scarce resources to the districts where they cut the most risk.
- **One shared truth:** ministries, NGOs and donors plan from the same live picture instead of competing reports.

NATIONAL ALIGNMENT

HOW HOZI ADVANCES THE NATIONAL AI STRATEGY

- Delivers the Strategy's **Year-1 Food Security** priority.
- Builds the "**AI disaster-warning system**" the Strategy explicitly calls for.
- Supports **Pillar 3** (AI for service delivery & decision-making).
- Acts as an analytics layer for the national data platform (**Project Pangolin**).
- Embodies **sovereignty, local language and responsible AI**.

WHY US

A RARE BLEND

Curious Inq pairs senior design and storytelling with applied AI. Most technical entries are powerful but hard to read; most beautiful entries are thin. Hoji is both — a serious engine wrapped in an interface a Minister can understand in thirty seconds.

Stage	Owner
Product, design, narrative	Nash Barara
Engine & deployment	[Technical Lead]
Domain validation	[Food-Security Advisor]

ROADMAP

Now (prototype): working engine + national map + simulator on sample data. **3 months:** live rainfall/satellite feeds, all staple crops, pilot with one ministry or NGO. **12 months:** national roll-out, local-language alerts to district officers, integration with the national data platform.

Hoji does not replace the experts. It gives them a head start — turning scattered numbers into a clear picture, early enough to act, so fewer families go hungry.